

Operator Discovery Report: scaffold_selector_transfer_focu S

- Operator type: scaffold_selector.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Selects a baseline TSP scaffold using instance descriptors: emphasize two-cluster bottlenecks and nearest-neighbor traps, then fall back to nearest-neighbor 2-opt. Helps held-out transfer without changing local-search in
- Novelty classification: descriptor_based_scaffold_selector.
- Rediscovery signature: scaffold_selector:random_restart_2opt:clustered_local_search:clustered_local_search.
- Surviving candidate: False.
- Problem class where it helps: grid_like_tsp.

Pseudocode

- Read instance descriptors to classify the layout family.
- Dispatch the instance to the preferred scaffold for that structure class.
- Fall back to a safe default scaffold when the structure is ambiguous.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: b7e4049e968c67d3.

Why It Should Help

- The operator selects among simple solvers from instance descriptors, so any gain should come from a reusable structure-to-scaffold mapping.

Failure Cases

- Underperforms on heldout_tsplib with gap delta 0.019828.
- Underperforms on elongated_corridor_tsp with gap delta 0.034174.
- Underperforms on nearest_neighbor_trap_tsp with gap delta 0.034212.
- Underperforms on two_cluster_bottleneck_tsp with gap delta 0.022965.
- Underperforms on uniform_euclidean with gap delta 0.013605.
- Does not transplant cleanly into nearest_neighbor_2opt.
- Does not transplant cleanly into cheapest_insertion_2opt.
- Does not transplant cleanly into random_restart_2opt.
- Does not transplant cleanly into sparse_three_opt.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.
- Runtime inflation is high at 0.682188.

Transplant Results

- nearest_neighbor_2opt gap delta 0.017077, runtime delta 201.277643, distance-eval delta 10421.285714.
- cheapest_insertion_2opt gap delta 0.124398, runtime delta -4254.496743, distance-eval delta -6750636.714286.
- random_restart_2opt gap delta 0.037402, runtime delta -100.886729, distance-eval delta -4254.0.
- sparse_three_opt gap delta 0.018362, runtime delta 216.8182, distance-eval delta 7000.0.
- clustered_local_search gap delta -0.007256, runtime delta 15.7217, distance-eval delta -4532.714286.

Instance-Family Results

- heldout_tsplib operator gap 0.254886 vs baseline 0.235058.
- clustered_tsp operator gap 0.113212 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.037378 vs baseline 0.003204.
- grid_like_tsp operator gap 0.082744 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.034212 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.132022 vs baseline 0.109057.
- uniform_euclidean operator gap 0.013605 vs baseline 0.0.

Pareto Results

- Gap delta: 0.017888.
- Runtime inflation: 0.682188.
- Distance-evaluation inflation: 1.485397.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator -0.0182.
- flattened_selector mean gap 0.121317 and delta vs full operator -0.0182.
- shuffled_selector mean gap 0.108754 and delta vs full operator -0.030763.